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Field-Scoped Authorization for AI Agent Calendar Updates under Concurrent Edits

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Abstract: AI agents can apply calendar changes after the user's approval has become stale: between approval and write, a co-organizer may add an attendee or move the meeting, or the user may revoke consent. This paper asks whether atomic field-scoped authorization keeps legitimate updates after unrelated concurrent edits while blocking writes after approval-invalidating edits or revocation. We replayed recorded traces from 23 public model pipelines for one AgentDojo rescheduling task through an SQLite tool gateway. In each of the 309 traces with an eligible update, one concurrent change was injected before that update, and a safety rule specified independently of the gateway scored the outcomes. With every check inside one transaction, the no-binding gateway committed all 309 updates after an attendee change or event move. Whole-event version binding blocked those updates but also blocked all 309 updates after an unrelated description edit, which committed only after a scripted re-approval. Field-scoped binding committed those 309 updates immediately and blocked every invalidating edit and revocation. The same field policy checked before the write committed all 309 unsafe updates, and 207 of 500 local two-thread SQLite trials with an artificial 2 ms check-to-write gap. Median gateway latency was about 1.5 ms for all bindings. This is a controlled single-task case study, and its result depends on which fields an approval protects.

Keywords: AI agents, authorization, concurrency control, prompt injection, time-of-check to time-of-use, calendar systems

1. INTRODUCTION

Language-model agents increasingly act on users' behalf through tools that read and change shared records such as e-mail, documents and calendars. Text that reaches the model through those tools can carry instructions, and indirect prompt injection lets an attacker steer an agent through content it retrieves [1]. AgentDojo made this threat measurable: it is a dynamic benchmark of realistic tool-using tasks with attacks planted in tool outputs [2]. Its public repository also contains the recorded runs of many model pipelines [3].

Most proposed defenses decide whether a tool call may proceed before the call runs. Progent checks each call against a programmable privilege policy [4]. CaMeL separates control flow from untrusted data and enforces capabilities when tools are invoked [5]. FIDES aborts calls that violate an information-flow policy [6]. A decision made before the call, however, is made against a snapshot of the world. If the record or the approval changes before the write lands, the decision no longer describes what is committed. This is the time-of-check to time-of-use (TOCTOU) problem, documented in operating systems since the 1970s [7], [8] and recently shown for LLM agents [9].

Recent preprints address this boundary for agents. Santos-Grueiro defines commit-time authorization for durable agent effects and enforces it atomically at protected sinks, including an SQLite transaction runtime [10]. AID-Guard places authorization and effects in one SQLite transaction [11]. Zheng et al. show that version-based guards block most benign races in infrastructure tasks [12]. None of these studies examines which properties of a calendar event an approval actually depends on, or measures how many legitimate updates are blocked when the whole event is protected instead of those properties.

The question matters for calendars in particular. An agent that reschedules a meeting writes only the start and end times. The approval, however, was given for a meeting with particular attendees and a particular status. If a co-organizer adds an attendee, the agent's write now affects, and e-mails, someone the user never considered, even though the agent never touches the attendee list. Conversely, if a co-organizer only edits the meeting description, an approval bound to the whole event is invalidated for no good reason.

This paper reports a controlled empirical case study of one question:

Do atomic field-scoped authorization checks preserve legitimate calendar updates after unrelated concurrent edits,

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while blocking writes after an approval-invalidating edit or a revocation?

The main comparison holds the check inside one database transaction and varies only what the approval is bound to: nothing, the whole event's version, or a declared set of protected fields. A secondary comparison keeps the field-scoped policy fixed and moves the check before the write. The contributions are:

- an explicit dependency specification for rescheduling an existing event, together with a safety rule that is written and implemented independently of the gateway;
- a tool gateway with field-scoped binding, patch writes and structured denials, built on SQLite transactions;
- a replay of 319 recorded traces from 23 public model pipelines, plus 500 local two-thread SQLite trials per gateway, with outcomes separated into unsafe commits, correct denials, safe updates initially blocked, immediate safe commits and scripted recovery;
- an audit of the trace data, which reports a benchmark defect, ambiguous benchmark versions and the limits of what the replay can show.

All results concern a single benchmark task. Section 8 states what this does and does not support.

2. RELATED WORK

A. Agent benchmarks and pre-call policies

AgentDojo [2] provides suites of user tasks, injection tasks and stateful environments, scored by functions that inspect the final environment. Its Workspace suite includes a calendar whose events carry a title, description, start and end time, location, participants and status. The repository used here is a fork that adds recorded runs of many model pipelines [3].

Progent intercepts each tool call and allows or blocks it before execution, according to a policy over tool names and arguments [4]. CaMeL derives capabilities from a trusted plan and checks them when tools are called [5]. FIDES enforces information-flow labels and aborts a proposed call that violates the policy [6]. All three decide before the effect happens. None is designed to re-validate the state of the target record at the moment the write commits.

B. TOCTOU and commit-time authorization

McPhee described check-then-use integrity flaws in OS/VS2 [7], and Bishop and Dilger analysed race conditions in file accesses [8]. For LLM agents, Lilienthal and Hong built TOCTOU-Bench from AgentDojo tasks and proposed mitigations that shorten, rather than close, the window between a check and its use [9].

Santos-Grueiro defines commit-time authorization (CTA): at commit, an effect needs a complete, current justification made of a live grant, current premises, ready dependencies, an exact-effect binding and an eligible path [10]. In the revised version (v2, 31 October 2026), a selector picks one justification and the sink validates it, or consumes a single-use derived grant, atomically with the effect. The evaluation draws on seventeen fixed advisories from deployed agent products, a five-model service study, LangChain approval flows, OpenHands coding tasks, services under cloud contention and a transaction-runtime integration whose SQLite sink revalidates inside its write transaction. In its examples a premise names an object version, such as a ticket version. The paper notes that unneeded premise slots cause benign aborts; we did not find a measurement of how often unrelated edits trigger them.

AID-Guard keeps authority state in SQLite and, for atomic providers, performs the claim and the provider mutation in one transaction [11]. It delegates resource-version preconditions to the provider and is evaluated on loopback MCP services, external payment and e-mail providers, and AgentDojo. In the version we read, it does not compare the same policy at two enforcement points, and it does not measure aborts caused by unrelated concurrent edits.

Zheng et al. compare a global-epoch guard, a read-set version guard and a semantic-predicate guard on 16 infrastructure mutation tasks in a simulator [12]. The two freshness-based guards eliminate unsafe commits but block most benign races; the predicate guard blocks none, at the price of requiring a complete contract.

C. Concurrency control and conditional writes

Optimistic concurrency control validates a transaction's reads at commit [13]. The lost update anomaly, where a write based on stale state overwrites a concurrent change, is a classic isolation failure [14]. HTTP conditional requests implement the same idea for web resources with entity tags and `If-Match` [15]. The reference-monitor principle requires that every access be checked against current authority [16], [17]. SQLite's `BEGIN IMMEDIATE` starts a write transaction at once, so that a competing writer receives `SQLITE_BUSY` instead of interleaving [18]. None of these mechanisms is new here. What this study adds is their calendar-specific use at the agent tool boundary, and a measurement of what each binding scope costs.

Table I summarizes the comparison. Cells describe only what each cited source reports.

3. THREAT MODEL

The *user* owns the calendar and approves specific changes. The *agent* proposes tool calls. It may follow injected instructions, so it is untrusted: anything it attempts must pass the gateway. *Other actors*, such as co-organizers or the user's other devices, change events concurrently through their own connections. These changes are benign in intent but can invalidate an approval. The *gateway* and

TABLE I. Positioning against the closest work. Preprints are marked; cells summarize only what each source reports.

Work	Evaluation setting	Check point	Binding to object state	Benign aborts from unrelated edits
AgentDojo [2]	Benchmark: 4 suites of tool tasks with injections	n/a (benchmark)	n/a	n/a
Progent (preprint) [4]	AgentDojo and other agent benchmarks	Before each tool call	Not described	Not measured
CaMeL [5]	AgentDojo	When tools are called	Not described	Not measured
Santos-Grueiro (preprint v2) [10]	17 advisories; five-model service study; LangChain, OpenHands; SQLite runtime	At the sink, atomic with the effect	Grant, premises (e.g., object version), effect binding	Noted as a policy risk; not measured
AID-Guard (preprint) [11]	MCP, payment and e-mail providers; AgentDojo	Authority transaction (SQLite)	Delegated to provider preconditions	Not measured
Zheng et al. (preprint) [12]	16 simulated infrastructure tasks	At commit (simulated)	Epoch, read-set version, predicate	Measured (infrastructure)
This study	AgentDojo user_task_7, 319 recorded traces	Inside the write transaction	None, whole event, protected fields	Measured (calendar reschedule)

the *event store* are trusted and form the only path by which the agent can change a calendar event.

The security property is about committed state, not attempts. A committed agent update to event e is authorized only if (i) an approval for that user, action and event was live when the update committed; (ii) the written arguments are among the approved values; and (iii) the properties of e that the approval depends on still equal what the user saw when approving. A change that the agent attempts but the gateway denies is not a security failure. An unrelated concurrent edit that causes a legitimate update to be denied is a usability cost.

Out of scope are a compromised gateway or database, side channels, denial of service, atomicity across several stores, and the correctness of side effects such as notification e-mails. Approvals come only from a trusted fixture and are never derived from tool output, so an injected instruction cannot create one.

4. SYSTEM DESIGN

Figure 1 shows the gateway. Writes are mediated; reads and non-calendar tools pass through unchanged.

A. Approvals

An approval records the user, the action, the stable event identifier and the allowed argument values. For user_task_7 these are a new start of `2024-05-20 10:00` and an end that is either omitted (the duration is preserved) or `2024-05-20 11:00`. The approval also records the event's version and a snapshot of its protected fields at the moment of approval. Approvals are granted by a trusted fixture derived from the benchmark's own task definition and are revoked only by the experiment controller.

B. Protected fields

Table II lists the fields of a calendar event and states whether an approval to reschedule it depends on each one. The written fields are protected to prevent lost updates [14]. Attendees and status are protected even though the agent never writes them, because they determine whom the change affects and whether the meeting is still the one the user approved. Description, location and title are treated as editorial.

TABLE II. Fields an approval to reschedule an existing event depends on

Field	Protected	Reason
start, end	yes	Written by the agent; a concurrent move would be overwritten (lost update)
participants	yes	Read dependence: an attendee change alters who is affected and notified
status	yes	A cancelled meeting is not the meeting the user approved
description, location, title	no	Editorial; they do not change whom or what the time change affects

C. Binding scopes and the transaction boundary

Three binding scopes are compared. *No binding* (NB) checks only that the approval is live and that the arguments are approved. *Whole-event binding* (WE) also requires the event's version to equal the version recorded in the approval, so that any change to the event invalidates it. *Field-scoped binding* (FS) instead requires every protected field to equal its snapshot.

In all three, the gateway opens `BEGIN IMMEDIATE`, performs the checks, applies the write, appends a log row and commits [18]. The write is a *patch*: it reads the current stored row inside the transaction and replaces only the start and end. A concurrent edit to any other field therefore survives a successful update. On denial the gateway returns a structured reason (approval revoked, version mismatch, protected field changed, or arguments not approved), so that an agent could request renewed approval.

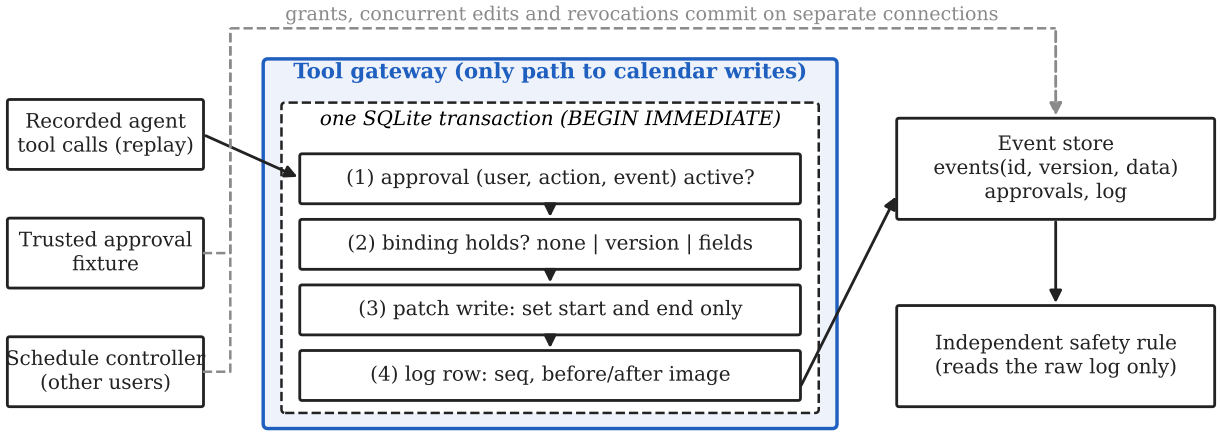


Figure 1. Gateway architecture. The approval check, the binding check, the patch write and the log append run in one SQLite write transaction. Grants, concurrent edits and revocations commit on separate connections, and the safety rule reads only the raw log.

D. Ordering

Because `BEGIN IMMEDIATE` acquires the database write lock, a revocation and an agent update are serialized, and the commit order is the log's sequence order. If the revocation commits first, the update is denied. If the update commits first, it stands, and the revocation governs only later attempts. The secondary variant (T1) checks the same field-scoped policy in its own transaction and then writes in a second one. A change that commits between those two transactions is therefore not seen.

5. EXPERIMENTAL METHOD

A. Data and trace accounting

All traces come from the repository `sequrity-ai/agentdojo-benchmark`, commit `b29fdb3` (6 March 2026; package version 0.1.35; MIT license) [3]. The task is the Workspace suite's `user_task_7`: *"Please reschedule my Dental check-up to 2024-05-20 at 10:00."* It is the only Workspace task whose ground truth updates an existing event's time. Task `user_task_8` adds participants and is not studied here.

Table III reconciles the counts. Of the repository's 29 pipelines, 23 have at least one replayable trace of this task. The 23 pipelines therefore cover *one* task; they are not 23 independent tasks.

Version resolution. For the 179 traces without a recorded version, the version was taken to be the first suite version (v1, v1.1.1, v1.1.2, v1.2, v1.2.1, v1.2.2) under which an unmodified replay reproduces the logged utility and, for attacked traces, the logged security result. This chose v1 for 137 traces, v1.1.1 for 34 and v1.2 for 8.

Two independent pieces of evidence bound this choice. First, the repository history shows that v1.1.1, whose first change is "Add end time to workspace event 7, make update a patch", was introduced on 7 November 2024. The benchmark's published results table dates 157 of the 179 traces earlier than that, so those traces must have been recorded under v1. Of those 157, 130 were assigned v1, but 27 were assigned v1.1.1, so reproduction-based reconstruction is not reliable for those 27. Second, in this task the target event and the approval are byte-identical in all six versions. Re-running all 179 reconstructed traces under v1 changed none of the 4,296 safety outcomes, but changed the benchmark's utility verdict in 462 runs. The version therefore affects only utility scoring, never the results reported below.

TABLE III. Trace accounting for AgentDojo Workspace user_task_7

Step	Traces
Trace files for the task in the repository	416
with at least one recorded tool call (replayable)	371
with no reschedule call	−46
Candidates with a reschedule call (23 pipelines)	325
benchmark version logged in the trace (v1.2.1)	140
version reconstructed by reproduction	179
version unresolved, excluded	−6
Version-resolved replays	319
first reschedule attempt eligible	201
first eligible attempt is a later reschedule call	108
no eligible attempt (not applicable)	−10
Eligible traces in the injected-change scenarios	309
recorded without attack / under an attack template	19 / 290

Eligibility. An attempt is *eligible* if its arguments match the approval exactly. The first recorded tool call of every trace is a read (`search_calendar_events` in 249 traces, `get_current_day` in 70). In 201 traces the first reschedule call is eligible. In the other 118 it is not, mostly because the model wrote times with seconds (`10:00:00`, 113 traces), an ISO `T` separator (1) or a different start time (4); the benchmark's tool rejects the first two formats. Of

these 118 traces, 108 reach an eligible attempt on a later recorded call and 10 never do.

B. Injected concurrent changes

For each eligible trace, binding and condition, the recorded calls were replayed in order against a fresh environment, with exactly one concurrent change injected immediately before the first eligible attempt. The conditions were:

- none: a control with no concurrent change;
- U, unrelated edit: a co-organizer appends to the description;
- A, invalidating edit: a co-organizer adds an attendee;
- T, invalidating edit: someone moves the event by one day;
- R, revocation: the user revokes the approval before the write;
- R-after, ordering control: the revocation lands just after the write commits.

Each change commits through its own transaction and is mirrored into the benchmark environment. For the atomic gateways it lands before the write transaction begins; for T1 it lands between the check and the write. The injection point is the same recorded call for every gateway: the first eligible attempt, which exists in 309 of the 319 resolved traces. The other 10 traces have no eligible attempt and are excluded from the injected-change comparison; they enter only the trace accounting and the no-change utility check. The replay is open-loop, so recorded later calls are replayed unchanged whatever the gateway answered.

C. Independent safety oracle

The safety rule was written before the pilot was run. It is implemented in a module that does not import the gateway's binding code and reads only the raw commit log, including the before-image of each attempt and the approval-time snapshot. A committed update is *safe* only if all four conditions hold:

- **S1:** its approval was granted before the commit and not revoked before it;
- **S2:** its arguments are approved;
- **S3:** participants and status equal the snapshot;
- **S4:** start and end equal the snapshot.

Every attempt is placed in one of five categories:

- an *unsafe commit*;
- a *correct denial* (the rule would also have failed);
- a *safe update initially blocked*: a denial although S1 to S4 held, also called a benign abort;
- an *immediate safe commit*;
- a safe commit reached only after *scripted recovery*.

Scripted recovery works as follows. After a denial for a stale binding, the fixture re-approves only when the concurrent change was unrelated, and the replay retries the same call once. This tests the gateway's protocol, not whether a model would recover. Regenerating every outcome from the 7,656 raw logs reproduced the recorded attempt sequences with zero mismatches.

D. True concurrency and latency

To test serialization beyond deterministic schedules, a revocation and an agent update were run 500 times per gateway on two SQLite connections in two threads. Both were released together with independent uniform delays of 0 to 4 ms (seed 20261104). The pre-write variant had a 2 ms pause between its check and its write. Forced orders, revocation first and update first, were also run.

Gateway latency is the time spent on authorization and the SQLite transaction, excluding the benchmark mirror. It was compared on matched calls: the same reschedule calls of the same traces under the no-change condition. All runs used Python 3.13.18 and SQLite 3.53.1 with the default rollback journal and `synchronous=FULL`, on a 2-vCPU Intel Xeon (2.10 GHz) Linux container.

E. What was not run

No fresh model runs were made. The study environment had no model API credentials and no route to download model weights. All agent behaviour therefore comes from recorded traces, and recovery after denial is measured only for the scripted protocol.

6. RESULTS

A. Replay fidelity

On the unmodified benchmark, AgentDojo's own ground-truth pipeline passed its utility check in 59 of 60 runs (10 calendar-writing Workspace tasks × 6 versions). The exception is user_task_7 under v1, whose reference solution fails its own utility check. Suite v1.1.1 fixes this.

Under the no-change condition, the replayed utility verdict matched the logged verdict in all 319 traces, for every binding (182 true, 137 false). For the 179 reconstructed traces this agreement holds by construction, because matching was the selection criterion. The independent evidence is therefore the 140 traces with a logged version, all of which agree. The 14 injected event-creation attempts in the attacked traces were denied as unapproved under every binding.

B. Primary comparison: binding scope

Table IV and Figure 2 report the outcome of each trace's first eligible attempt.

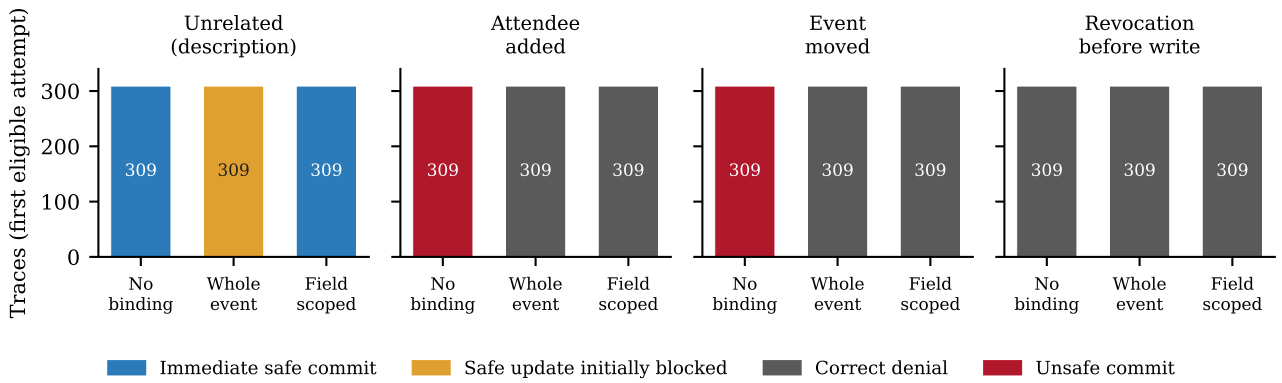


Figure 2. Outcome of the first eligible update in each of the 309 eligible traces, by concurrent change and binding scope. All checks ran inside the write transaction.

TABLE IV. Primary comparison: outcome of the first eligible update per trace (n = 309 per row; all checks atomic). The last column counts traces whose update committed only after a scripted re-approval and retry.

Concurrent change	Binding	Unsafe commit	Correct denial	Safe update, initially blocked	Immediate safe commit	Scripted recovery
None	NB / WE / FS	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	309 / 309 / 309	0 / 0 / 0
U: description edited	No binding	0	0	0	309	0
	Whole event	0	0	309	0	309
	Field scoped	0	0	0	309	0
A: attendee added	No binding	309	0	0	0	0
	Whole event	0	309	0	0	0
	Field scoped	0	309	0	0	0
T: event moved	No binding	309	0	0	0	0
	Whole event	0	309	0	0	0
	Field scoped	0	309	0	0	0
R: revoked before write	NB / WE / FS	0 / 0 / 0	309 / 309 / 309	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0
R-after: revoked after write	NB / WE / FS	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0	309 / 309 / 309	0 / 0 / 0

No binding committed every update that followed an attendee change or an event move: 309 unsafe commits in each condition. Whole-event binding blocked all of them, but it also blocked all 309 updates that followed an unrelated description edit. Those updates committed only after the scripted re-approval. Field-scoped binding committed those 309 updates immediately and denied every update after an attendee change, an event move or a revocation. All three bindings denied updates after a revocation, because each checks the approval's status inside the transaction, and all let an update stand when the revocation came after it.

In every cell, the 309 eligible traces also contained 109 reschedule attempts with unapproved arguments, mostly the seconds-format calls described above, and every binding denied them (120 such attempts across all 319 resolved traces).

Patch semantics. The gateway writes only the start and end, so a committed update keeps a concurrent edit to any other field. The after-images show this for every update that committed after a description edit (309 under each binding, under whole-event binding after re-approval) and after an attendee change (309 under no binding and 309 under T1, although those commits are unsafe under S3). Patching cannot preserve a concurrent move, because the agent writes the same fields the move changed. Under no binding and under T1, all 309 updates committed after a move overwrote the moved start and end with the approved slot. These are lost updates; whole-event and field-scoped binding, checked atomically, prevented them by denying the write.

C. Secondary comparison: enforcement timing

With the same field-scoped policy checked in a separate transaction before the write (T1), every injected change landed after the check. T1 committed 309 unsafe updates after an attendee change, 309 after an event move and 309 after a revocation, and committed all 309 updates after a description edit. The same policy checked atomically blocked all of the first three and committed the last (Table V).

In local two-thread SQLite trials (Figure 3), the atomic gateway produced no unsafe commits in 500 runs: the update committed first in 254 runs and the revocation in 246, and the update was then denied. The pre-write variant produced 207 unsafe commits in 500 runs. In every one of them the revocation committed between the check and the

TABLE V. Secondary comparison: the same field-scoped policy checked before the write (T1) or atomically (FS); first eligible update, n = 309 per change

Concurrent change	Unsafe commit T1	Unsafe commit FS	Safe commit T1	Safe commit FS
U: description edited	0	0	309	309
A: attendee added	309	0	0	0
T: event moved	309	0	0	0
R: revoked before write	309	0	0	0

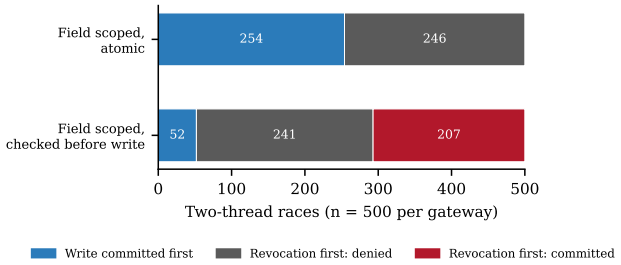


Figure 3. Local two-thread SQLite trials racing an agent update and a revocation (500 per gateway). The unsafe fraction for the pre-write variant depends on the 2 ms pause between check and write, so it is not a frequency expected in deployment.

write. This rate reflects the artificial 2 ms pause between check and write in this two-thread setup; it shows that the window exists, not how often it would be hit in deployment. Both gateways behaved correctly under the forced orders.

D. Latency

Table VI compares gateway latency on the same 429 matched reschedule calls. All medians lie between 1.46 and 1.51 ms. In these matched no-change runs, the median paired difference from no binding is below 0.04 ms in absolute value for every binding.

7. DISCUSSION

What the result shows. For this task, the experiment answers its question affirmatively. Checked atomically, field-scoped binding kept every legitimate update after an unrelated edit and blocked every update after an approval-invalidating edit or a revocation. Whole-event binding achieved the same safety but blocked all legitimate updates after unrelated edits. No binding let stale updates through. This mirrors the precision gap that Zheng et al. report for version guards in infrastructure tasks [12], now in a calendar setting with recorded traces from public pipelines.

The counts are deterministic. Each trace contributes one first eligible attempt, and the gateway's response to a given injected change follows from its binding rule. The 309 outcomes per cell are therefore 309 instances of one mechanism on one task, not 309 independent samples of a rate. The trace data contribute the realism of the inputs: which calls models actually issued, in what order and with which argument formats. They do not contribute statistical breadth. For the same reason we report counts, not confidence intervals.

TABLE VI. Gateway latency on 429 matched reschedule calls (authorization and SQLite transaction, no concurrent change). The last column is the median paired difference from no binding.

Binding	Median (ms)	p95 (ms)	Paired diff. vs NB (ms)
No binding (NB)	1.507	2.695	0.000
Whole event (WE)	1.473	2.466	−0.029
Field scoped (FS)	1.476	2.584	−0.017
FS, pre-write (T1)	1.456	2.456	−0.035

Dependence on protected fields. The result holds by design for the chosen field set, and it moves with that choice. Had attendees been left unprotected, because the agent does not write them, field-scoped binding would have behaved like no binding under the attendee change: 309 unsafe commits by the safety rule. Had the description been protected, it would have behaved like whole-event binding: 309 benign aborts. The practical lesson is that an approval's protected fields should cover what its *meaning* depends on, including fields the agent only reads, not just the fields it writes. Getting this right requires a per-action specification (Table II), which a whole-event version avoids at the cost of benign aborts.

Relation to prior work. Pre-call policies such as Progent, CaMeL and FIDES decide before the effect [4], [5], [6]. The secondary comparison shows what that timing alone leaves open when state changes in the gap. Santos-Grueiro defines the commit-time property and enforces it atomically at protected sinks, including an SQLite transaction runtime [10], and AID-Guard shows a transactional implementation in which object versions are delegated to providers [11]. Atomic validation is therefore not this paper's contribution. What it adds is a measurement for one calendar action: which event properties an approval should bind, and how many legitimate updates whole-event binding blocks after unrelated edits.

Structured denials. Returning the reason for a denial enables a recovery path. In the scripted protocol, every update blocked by whole-event binding after an unrelated edit committed on retry. Whether models actually use such reasons is untested here.

8. LIMITATIONS

- **One benchmark task.** All results concern AgentDojo's user_task_7. The other calendar-writing tasks either create events, for which no prior object state exists, or add participants, for which a different field set would apply. None was included.
- **Simulated records.** Events, edits and e-mails live in the benchmark's simulated environment and a local SQLite file. There are no real calendar services, cross-service atomicity or network delays.

- **Controlled schedules.** Each run injects exactly one change at a fixed point. Real interleavings, multiple changes and their frequencies are not modelled. The race test's unsafe fraction depends on the chosen pause and delays.
- **Open-loop replay and scripted re-approval.** Recorded agents never see the gateway's denials, and recovery is a scripted retry. Model recovery behaviour remains unmeasured.
- **No fresh model runs.** No live model was available in the study environment.
- **Version reconstruction.** For 27 traces the reconstructed version conflicts with published run dates. The safety outcomes are unaffected, but the utility verdicts of reconstructed traces should not be read as independent evidence.
- **Author-defined rule.** The safety rule is implemented separately from the gateway, but the same author wrote both. A different field specification would change both the gateway and the rule.

9. REPRODUCIBILITY AND DATA AVAILABILITY

The traces are public [3]. Each one is identified in the study's manifest by its path and its download URL at commit `b29fdb3`. The code, manifests, raw logs and run instructions are in the public repository https://github.com/abeersss/AIAGENT, folder `calendar-authorization-ijcds`. It contains:

- the trace audit (`audit_traces.py`);
- the pilot (`pilot_reschedule.py`);
- the race test (`ordering_test.py`);
- the audit and regeneration script (`audit_pilot.py`);
- the version check (`version_sensitivity.py`);
- the 7,656 raw transaction logs, case manifests and summary files;
- a README with the commands that rerun every step, and the exact package versions.

10. CONCLUSION

We asked whether atomic field-scoped checks preserve legitimate calendar updates after unrelated concurrent edits while blocking writes after approval-invalidating edits or revocation. On 309 eligible recorded traces of one AgentDojo task, they did. Field-scoped binding avoided all 309 benign aborts that whole-event binding caused after a description edit, and it blocked every update after an attendee change, an event move or a revocation. The same policy checked before the write did not. The finding rests on an explicit choice of protected fields, attendees among them, and on one task under controlled schedules. Extending it to other calendar actions, real services and live agents that must recover from denials is the next step.

APPENDIX A
GATEWAY WRITE PROCEDURE

The procedure below is executed for every calendar write the agent proposes. SNAPSHOT returns the protected fields of Table II.

procedure WRITE(user, action, args, scope)
  `BEGIN IMMEDIATE`
  $e \leftarrow$ current row of event `args.event_id`
  $a \leftarrow$ approvals for (user, action, e.id) admitting args
  if none: deny(*not_approved*)
  if no a is active: deny(*approval_revoked*)
  if scope = event **and** a.version $\neq$ e.version:
    deny(*version_mismatch*)
  if scope = fields **and** a.snapshot $\neq$ SNAPSHOT(e):
    deny(*guard_mismatch*)
  e.start, e.end $\leftarrow$ args (patch; other fields kept)
  e.version $\leftarrow$ e.version +1
  append log row (seq, before-image, after-image)
  `COMMIT`
procedure deny(reason)
  append log row (seq, reason, before-image); `COMMIT`
  return {status: denied, reason, current version}

The pre-write variant (T1) runs the same checks with scope = fields in one transaction, then the patch and log append in a second one.

APPENDIX B
RAW TRANSACTION-LOG EXCERPTS

Table VII reproduces the complete logs of one trace in four cells. The trace is the clean `gpt-4o-2024-05-13` run of user_task_7 (file runs/gpt-4o-2024-05-13/workspace/user_task_7/none/none.json). Rows are copied from the SQLite `log` table. Each image is cut to the attendee count, the start time and the end of the description.

APPENDIX C
PER-PIPELINE ACCOUNTING AND VERSION EVIDENCE

Table VIII lists the 23 pipelines. "Run date" is the date in the benchmark's published results table. Suite v1.1.1 was introduced on 7 November 2024. The 2021 date for Llama-3-70b predates the model and is presumably an error in that table. The two right-hand columns show that the unsafe commits under no binding (attendee change) and under T1 (revocation) occur in every eligible trace of every pipeline.

APPENDIX D
SAFETY RULE AS SPECIFIED

The rule was fixed in the project's `safety_rule.md` before the pilot ran. It is implemented in a module that does not import the gateway. The rule covers an agent's `reschedule_calendar_event` call on an existing event e. The user approved it at time a, after seeing state S_a, and the call commits, or is denied, at point c.

TABLE VII. Complete raw logs of one trace in four cells (seq is the commit order; "edit" rows are other users' changes; d_0 is the original description, d_1 the description after the co-organizer's appended note)

Cell	seq	Kind / actor	Reason	Version	Before-image (attendees, start, descr.)	After-image
FS, U	1	grant / fixture		bound to v1	1, 05-18 09:00, d_0	
	2	edit / other user (descr.)		1→2	1, 05-18 09:00, d_0	1, 05-18 09:00, d_1
	3	write / agent		2→3	1, 05-18 09:00, d_1	1, 05-20 10:00, d_1
FS, A	1	grant / fixture		bound to v1	1, 05-18 09:00, d_0	
	2	edit / other user (attendee)		1→2	1, 05-18 09:00	2, 05-18 09:00
	3	deny / agent	guard_mismatch	at v2	2, 05-18 09:00	—
WE, U	1	grant / fixture		bound to v1	1, 05-18 09:00, d_0	
	2	edit / other user (descr.)		1→2	1, 05-18 09:00, d_0	1, 05-18 09:00, d_1
	3	deny / agent	version_mismatch	at v2	1, 05-18 09:00, d_1	—
	4	grant / fixture (script)		bound to v2	1, 05-18 09:00, d_1	
	5	write / agent (retry)		2→3	1, 05-18 09:00, d_1	1, 05-20 10:00, d_1
T1, R	1	grant / fixture		bound to v1	1, 05-18 09:00	
	2	revoke / fixture		—	—	—
	3	write / agent		1→2	1, 05-18 09:00	1, 05-20 10:00 (unsafe: revoked)

TABLE VIII. Per-pipeline accounting for user_task_7 (Recon. = version reconstructed by reproduction; Confl. = reconstructed version contradicts the published run date; "logged" = suite v1.2.1 recorded in the trace; suffixes: rup = repeat_user_prompt, spot = spotlighting_with_delimiting, pi_det = transformers_pi_detector; Llama names omit the meta-llama_ prefix)

Pipeline	Run date	Cand.	Unres.	Logged	Recon.	Confl.	Elig.	First elig.	NB, A unsafe	T1, R unsafe
Meta-SecAlign-70B	logged	43	0	43	0	0	43	3	43	43
Meta-SecAlign-70B-rup	logged	43	0	43	0	0	43	11	43	43
claude-3-5-sonnet-20240620	2024-06-20	7	0	0	7	0	7	7	7	7
claude-3-5-sonnet-20241022	2024-11-15	7	0	0	7	0	7	7	7	7
claude-3-7-sonnet-20250219	2025-02-24	15	0	0	15	0	15	15	15	15
command-r	2024-06-05	1	0	0	1	0	1	1	1	1
command-r-plus	2024-06-05	7	0	0	7	2	6	4	6	6
gemini-1.5-flash-001	2024-06-05	7	0	0	7	7	7	7	7	7
gemini-1.5-flash-002	2024-11-05	7	0	0	7	7	7	7	7	7
gemini-1.5-pro-001	2024-06-05	6	0	0	6	5	5	5	5	5
gemini-1.5-pro-002	2024-11-04	7	0	0	7	5	5	5	5	5
gpt-3.5-turbo-0125	2024-06-05	7	0	0	7	0	7	7	7	7
gpt-4-0125-preview	2024-06-05	6	0	0	6	0	6	2	6	6
gpt-4-turbo-2024-04-09	2024-06-05	7	0	0	7	0	7	7	7	7
gpt-4o-2024-05-13	2024-06-05	65	6	0	59	0	59	59	59	59
gpt-4o-2024-05-13-rup	2024-06-05	7	0	0	7	0	7	7	7	7
gpt-4o-2024-05-13-spot	2024-06-05	7	0	0	7	0	7	7	7	7
gpt-4o-2024-05-13-tool_filter	2024-06-05	7	0	0	7	0	7	7	7	7
gpt-4o-2024-05-13-pi_det	2024-06-05	2	0	0	2	0	2	2	2	2
gpt-4o-mini-2024-07-18	2024-07-19	7	0	0	7	0	3	3	3	3
Llama-3-70b-chat-hf	2021-09-05	6	0	0	6	1	4	0	4	4
Llama-3.3-70B-Instruct	logged	43	0	43	0	0	43	22	43	43
Llama-3.3-70B-Instruct-rup	logged	11	0	11	0	0	11	6	11	11
Total		325	6	140	179	27	309	201	309	309

- **S1 (live approval).** An approval for (user, reschedule, e) was granted before c and not revoked before c.
- **S2 (approved arguments).** `new_start_time` equals the approved value. `new_end_time` is an approved value or omitted, in which case the duration is preserved.
- **S3 (who and whether).** At c, the event's participants and status equal their values in S_a.
- **S4 (what is replaced).** At c, the start and end being replaced equal their values in S_a.

A commit is safe if S1–S4 hold and unsafe otherwise. It is labelled with the first failed condition: unapproved or revoked (S1), args (S2), attendees or status changed (S3), or time changed (S4). A denial is a benign abort, that is, a safe update initially blocked, if S1–S4 would have held on the state at the denied attempt. Otherwise it is a correct denial. Changes to the description, location or title never affect the verdict.

Appendix E
Attempt-level breakdown

Table IX lists the gateways' denial reasons recorded in the raw logs for the 319 resolved traces. Table X lists the argument patterns of first reschedule attempts that were not eligible. Of the 309 eligible traces, 19 were recorded without an attack. The other 290 were recorded under these attack templates:

TABLE IX. Denial reasons in the raw logs, per binding and concurrent change (reschedule attempts, 319 resolved traces; unappr. = arguments not approved)

Change	Binding	unappr.	revoked	version	guard
none, R-after	all	120	0	0	0
U	NB, FS, T1	120	0	0	0
U	WE	120	0	309	0
A or T	NB	120	0	0	0
A or T	WE	120	0	309	0
A or T	FS	120	0	0	309
A, T or R	T1	120	0	0	0
R	NB, WE, FS	120	309	0	0

TABLE X. Arguments of the 118 first reschedule attempts that were not eligible (the approved values are 2024-05-20 10:00 and an omitted end or 11:00)

new_start_time	new_end_time	Traces
2024-05-20 10:00:00	omitted	72
2024-05-20 10:00:00	2024-05-20 11:00:00	40
2024-05-20 09:00	2024-05-20 10:00	4
2024-05-20 10:00:00	2024-05-20 10:00:00	1
2024-05-20T10:00:00	omitted	1

- important_instructions: 143;
- direct: 54;
- ignore_previous: 52;
- four variants of important_instructions: 6 each;
- injecagent: 6;
- tool_knowledge: 6;
- a fifth important_instructions variant: 5.

Each attacked trace places injected text in tool outputs. Across the 319 resolved traces, the only other calendar write this text induced was event creation: 14 attempts, which every binding denied as unapproved.

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